Evidence in cosmology: then and now

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BASICS OF BAYESIAN INFERENCE

Rev. Thomas Bayes (1701–1761)

- Collect a set of N data points D_i $(i = 1, 2, \ldots, N)$, which we denote collectively as the data vector $\boldsymbol{D}$.
- Propose some model (or hypothesis) H for the data, depending on M parameters θ_j $(j = 1, \ldots, M)$, that we denote by the parameter vector $\boldsymbol{\theta}$.
- Apply Bayes' theorem

$$\Pr(\boldsymbol{\theta}|\boldsymbol{D}, H) = \frac{\Pr(\boldsymbol{D}|\boldsymbol{\theta}, H)\Pr(\boldsymbol{\theta}|H)}{\Pr(\boldsymbol{D}|H)} \quad \rightarrow \quad P(\boldsymbol{\theta}) = \frac{L(\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{Z}$$

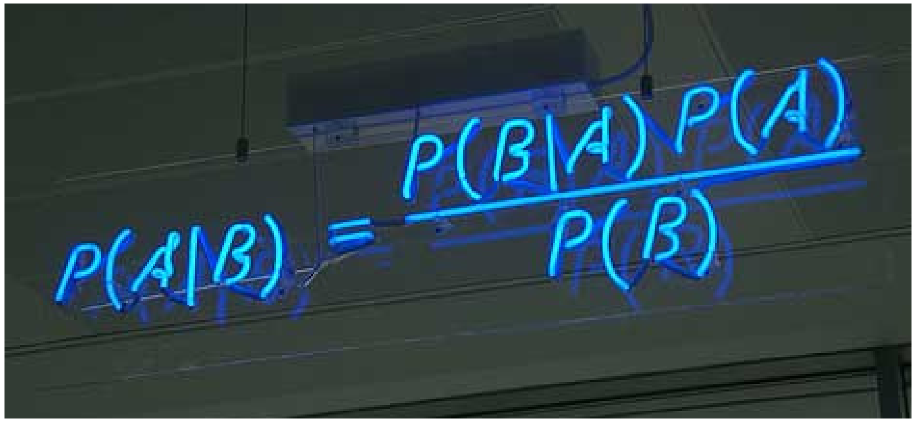
– prior $\pi(\boldsymbol{\theta}) \equiv \Pr(\boldsymbol{\theta}|H)$ represents our state of knowledge (or prejudices) of the parameter values before analysing the data

– likelihood $L(\boldsymbol{\theta}) \equiv \Pr(\boldsymbol{D}|\boldsymbol{\theta}, H)$ of the data given a particular set of parameter values, which modulates prior to give the...

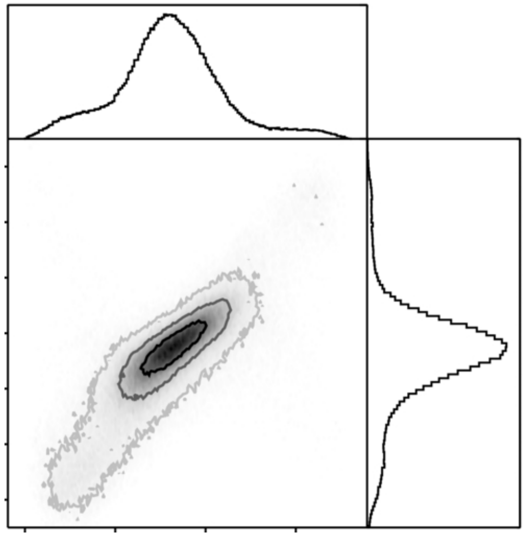
– posterior $P(\boldsymbol{\theta}) \equiv \Pr(\boldsymbol{\theta}|\boldsymbol{D}, H)$ which is the result, namely the complete inference

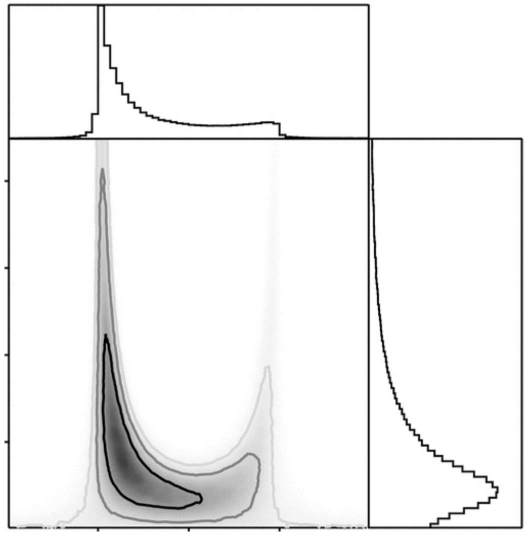
– evidence $Z \equiv \Pr(\boldsymbol{D}|H)$ provides normalisation of the posterior (and, as we see, it is much more!)

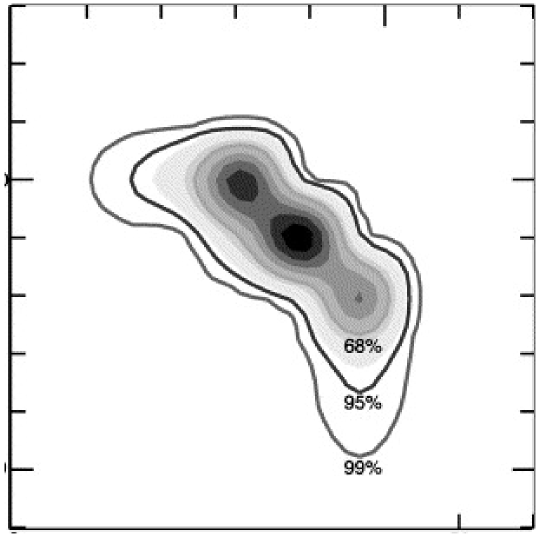
BAYESIAN PARAMETER ESTIMATION



- For parameter estimation, the posterior $P(\boldsymbol{\theta})$ is the complete inference
- Can summarise using peak(s) position(s) and covariance(s)
- Can obtain constraints on subsets of parameters by marginalisation







- Can maximise (typically non-convex) or, better, map out $P(\boldsymbol{\theta})$ using sampling

BAYESIAN MODEL SELECTION

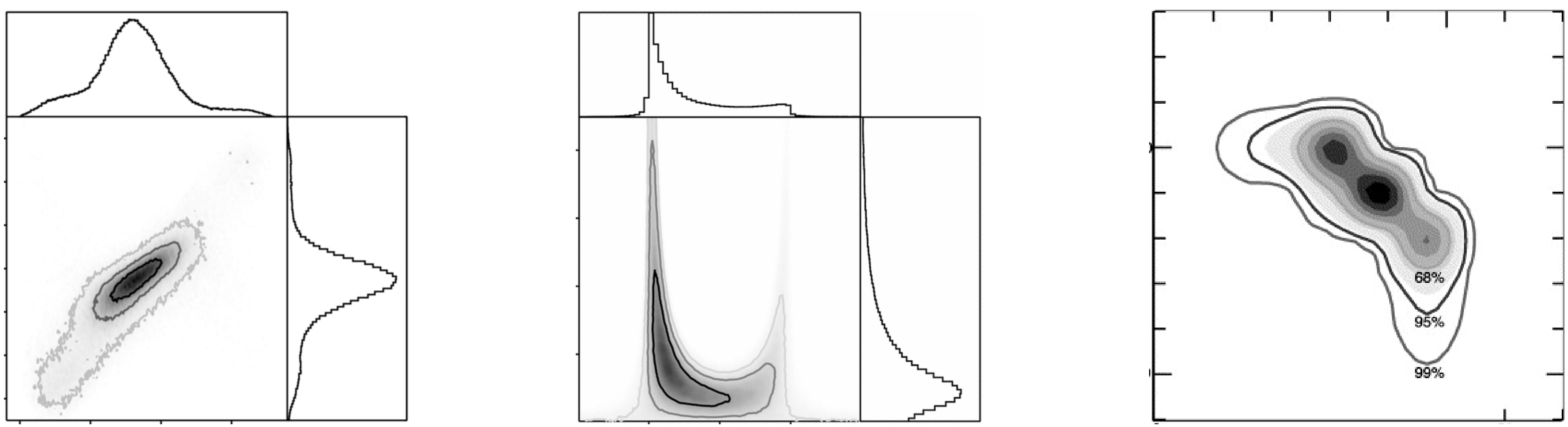
- Often wish to determine which of a set of alternative models best describes the data

- Model selection: for H_i $(i = 0, 1)$, the probability density associated with $\boldsymbol{D}$ is

$$Z_i \equiv \Pr(\boldsymbol{D}|H_i) = \int L_i(\boldsymbol{\theta})\pi_i(\boldsymbol{\theta})\, d\boldsymbol{\theta}$$

  Then consider posterior odds ratio $=$ bayes factor $+$ log prior ratio

$$\mathcal{P}_{10} \equiv \ln \frac{\Pr(H_1|\boldsymbol{D})}{\Pr(H_0|\boldsymbol{D})} = \ln \frac{Z_1}{Z_0} + \ln \frac{\Pr(H_1)}{\Pr(H_0)}$$



- Evidence naturally incorporates Occam's razor

- Calculating the evidence: analytic, Gaussian approximation(s), Savage-Dicke ratio, direct numerical integration, MCMC thermodynamic integration, nested sampling, . . .

THEN (30 YEARS AGO!): FIRST EVIDENCE APPLICATION IN COSMOLOGY

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H_0 AND ODDS ON COSMOLOGY

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ABSTRACT

Recent observations by the *Hubble Space Telescope* of Cepheids in the Virgo cluster imply a Hubble constant $H_0 = 80 \pm 17$ km s^{-1} Mpc^{-1}; other recent observations find $70 \lesssim H_0 \lesssim 90$ km s^{-1} Mpc^{-1}, with several large excursions in either direction. We attempt to clarify some issues of interpretation of these results for determining the global cosmological parameters Ω and Λ. In this paper, we use these results as a case study in the formalism of Bayesian model comparison, allowing a rigorous comparison of the different cosmological possibilities. We concentrate our analysis on three recent determinations of the Hubble constant, but the results are generic so long as they prefer $H_0 t_0 \gtrsim 1$, which would seem to require $\Lambda > 0$ within the context of Friedmann-Robertson-Walker cosmologies. With our more rigorous methods, the data do indeed suggest a universe with a nonzero cosmological constant but vanishing curvature: $\Omega + \Lambda = 1$.

Subject headings: cosmology: observations — cosmology: theory — distance scale — methods: statistical

- Jaffe (1996): which of four (Ω, Λ) models do the H_0 (and age) data prefer?

THEN: THE MEASUREMENTS AND THE MODEL

- Two independent (Gaussian) measurements:

  – Hubble constant $\widehat{H} \pm \sigma_H$ from HST, CFHT, SN Ia: $\mathrm{Pr}(\widehat{H} \,|\, H_0) = N(H_0, \widehat{H}, \sigma_H)$

  – oldest-object age (globular clusters) $\hat{t} \pm \sigma_t \approx 14.7 \pm 2$ Gyr: $\mathrm{Pr}(\hat{t} \,|\, \tau) = N(\tau, \hat{t}, \sigma_t)$

- Parameters $(H_0,\ t_0,\ \tau,\ \Omega,\ \Lambda)$ are joined by two relations:

  (i) *cosmology*: the dimensionless age depends only on (Ω, Λ)

$$H_0 t_0 = f(\Omega, \Lambda) = \int_0^1 \frac{da}{\sqrt{\underbrace{(1-\Omega-\Lambda)}_{\text{curvature}} + \underbrace{\Omega/a}_{\text{matter}} + \underbrace{\Lambda a^2}_{\Lambda}}}$$

  (ii) *astrophysics*: the Universe outlives its oldest stars, $t_0 > \tau$ — a one-sided bound (τ is the cluster age, not t_0!)

- Goal: constrain $(\Omega, \Lambda) \Rightarrow$ treat H_0, t_0, τ as nuisance parameters to integrate out

THEN: FROM MARGINALISATION TO MODEL SELECTION

- Integrate out the nuisance parameters in turn:
  - τ subject to $t_0 > \tau \;\Rightarrow\;$ a one-sided bound $t_0 \gtrsim \hat{t}$
  - H_0, t_0 subject to $H_0 t_0 = f \;\Rightarrow\;$ a likelihood in $f(\Omega, \Lambda)$ alone
- The result is an approximate lower limit on f — a cumulative normal Φ, not a peak:

$$L(\Omega,\Lambda) \approx \Phi\left(\frac{f(\Omega,\Lambda) - \widehat{H}\,\widehat{t}}{\sigma_f}\right), \qquad \sigma_f \simeq \hat{t}\,\sigma_H + \widehat{H}\,\sigma_t$$

- L increases monotonically with f: within any model the best fit runs to the edge ($\Omega \to 0$ or $\Lambda \to 1$) — parameter estimation alone is uninformative
- So the real question is which model? Four nested priors on (Ω, Λ) (0–2 parameters): (1) $\Omega = 1, \Lambda = 0$; (2) $\Omega + \Lambda = 1$; (3) $\Omega \le 1, \Lambda = 0$; (4) $\Omega, \Lambda \le 1$
- Compare by the evidence $Z_i = \int L_i(\boldsymbol{\theta})\,\pi_i(\boldsymbol{\theta})\,d\boldsymbol{\theta}$ (Gaussian approx. $+$ grid, log-uniform priors) — automatic Occam's razor

THEN: SIMPLE MODELS GIVE SIMPLE CONSTRAINTS

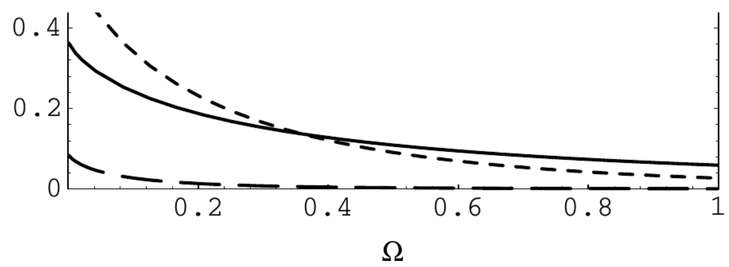


FIG. 1.—Posterior distribution for Ω for model 3: $\Omega \leq 1$, $\Lambda = 0$, given each of the three data sets, as labeled.

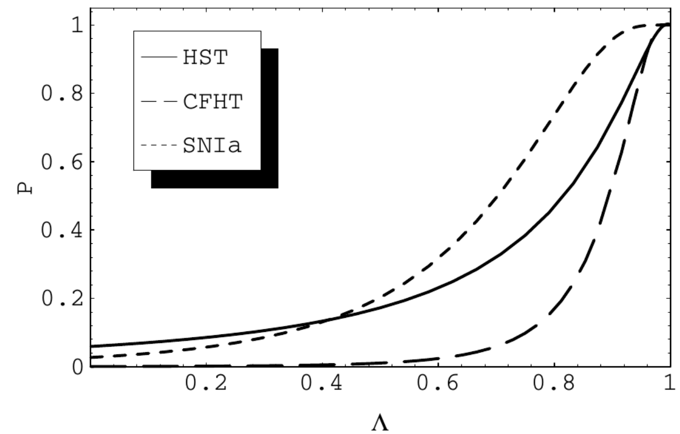


FIG. 2.—Posterior distribution for Λ for model 2: $\Omega + \Lambda = 1$, given ıch of the three data sets, as labeled.

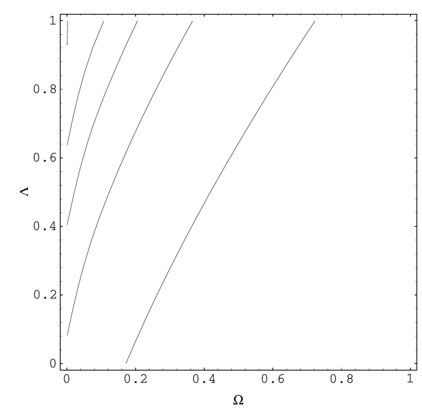


FIG. 3.—Posterior distribution for Ω and Λ for model 4: $\Omega, \Lambda \leq 1$, ıven the *HST* data, $H_0 = 80 \pm 17$ km s^{-1} Mpc^{-1}; contour spacing is 0.2 ıd the most likely value is the upper left corner ($\Lambda = 1, \Omega = 0$).

Posteriors for Ω (model 3), Λ (model 2), and the (Ω, Λ) plane (model 4)

- As anticipated, every posterior piles up against $\Omega \to 0$ or $\Lambda \to 1$ — the lower limit on f at work; the data parameter constraints alone cannot pick a model

THEN: THE EVIDENCE DELIVERS DECISIVE ODDS

TABLE 1

EVIDENCE FOR VARIOUS COSMOLOGICAL MODELS AND COMBINATIONS OF EXPERIMENTAL RESULTS, AS MARKED

HST (80 ± 17)	CFHT (80 ± 7)	SN Ia (66 ± 7)	$\Omega = 1$ ($\Lambda = 0$)	$\Omega \leq 1$ ($\Lambda = 0$)	$\Omega + \Lambda = 1$	$\Omega \leq 1$ ($\Lambda \leq 1$)	$\Omega + \Lambda \leq 1$	Average (1+2+3+4')
×			0.05902	0.1328	0.2835	0.235	0.1238	0.150
	×		0.0005454	0.009207	0.1316	0.06358	0.00548	0.0367
		×	0.02682	0.1413	0.3536	0.3062	0.1197	0.1604
×	×		0.00003219	0.002167	0.09948	0.03718	0.000911	0.0256
×		×	0.001583	0.02753	0.1835	0.1154	0.01772	0.0576
	×	×	0.00001463	0.00303	0.1186	0.04936	0.001071	0.0307
×	×	×	8.634×10^{-7}	0.0008212	0.09433	0.03221	0.0002092	0.0238
×	×	×	1.09×10^5:1	115:1	1:1	2.93:1	451:1	Odds

NOTE.—We also show the odds ratios for the combined data set.

- Odds are for the combined data, relative to flat $\Omega + \Lambda = 1$: favoured $115 : 1$ over open ($\Omega \leq 1, \Lambda = 0$) and $\sim 10^5 : 1$ over Einstein–de Sitter ($\Omega = 1$)

- Also favoured $451 : 1$ over the more general $\Omega + \Lambda \leq 1$ model that contains it — the unused freedom is penalised: Occam's razor, automatically

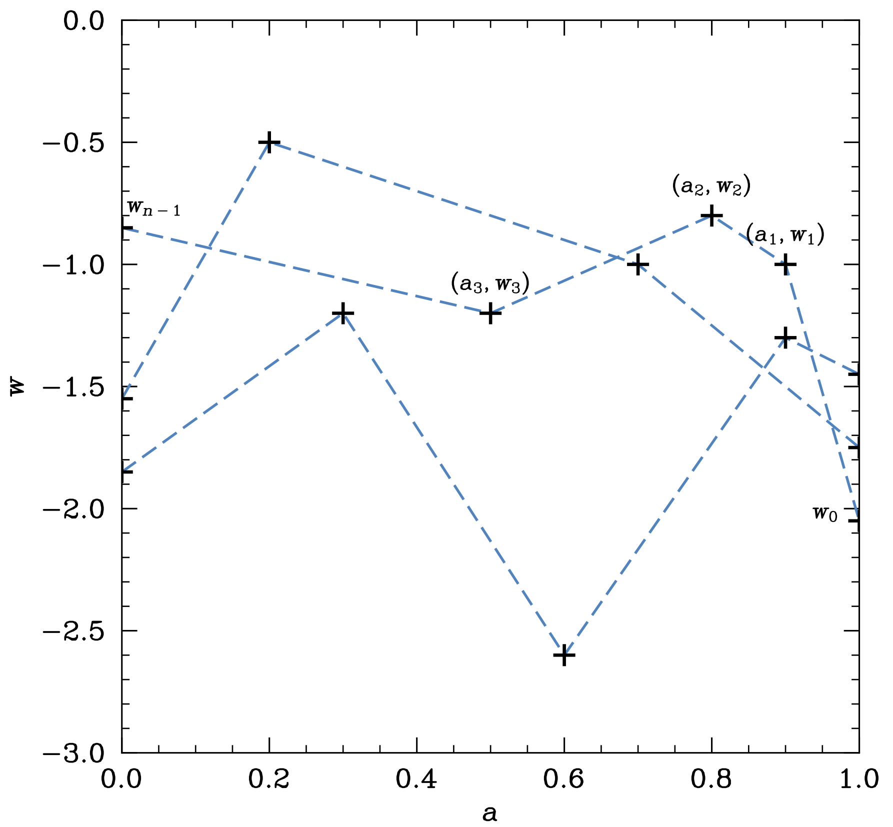
- The whole analysis: ≤ 2 parameters, Gaussian approximation $+$ grid integration

NOW: THE DARK ENERGY EQUATION OF STATE $w(z)$

- Is dark energy a cosmological constant ($w = -1$), or is it dynamical, $w(z)$?
- Data: DESI BAO (DR1, DR2) $+$ Pantheon+ / DES-5Y Type Ia supernovae
- Standard parametric answer: CPL form $w(a) = w_0 + (1-a)w_a$ (just 2 numbers)
- But why impose a fixed shape? The modern Bayesian toolkit lets the data choose:
  - nested sampling (PolyChord) computes the evidence in tens of dimensions
  - variable-dimension model averaging — the number of parameters is itself inferred
  - GPU acceleration — runs that took tens of minutes on CPU now take seconds
- Work by Adam Ormondroyd (Cambridge PhD, 2026), with Handley, Lasenby & MPH

NOW: FREE-FORM $w(a)$ WITH "FLEXKNOTS"

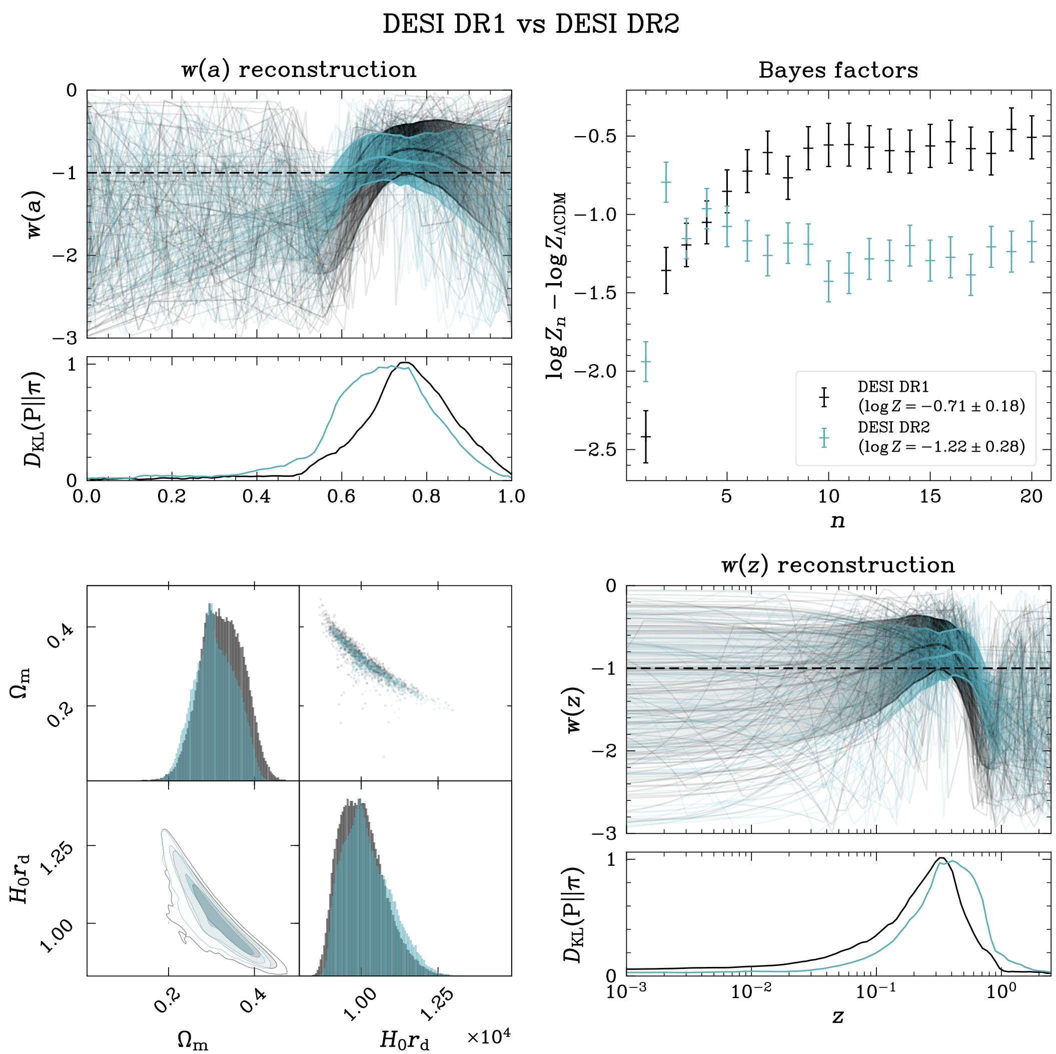
- Represent $w(a)$ as a linear spline between free-moving knots*; $w(z)$ follows directly from $a = 1/(1+z)$
- Interior knot positions and all values are free (end knots fixed at $a = 0, 1$)
- The number of knots n is itself a parameter:
  $n = 1$: wCDM (constant w)
  $n = 2$: CPL
  n up to 20: ~ 40-D, fully free-form
- PolyChord returns the evidence $Z(n)$ for each n — then marginalise over n
- No fixed functional form is imposed — the data sculpt $w(a)$



* First applied in cosmology to reconstruct the primordial power spectrum $P(k)$: Vázquez, Bridges, Hobson & Lasenby (2012)

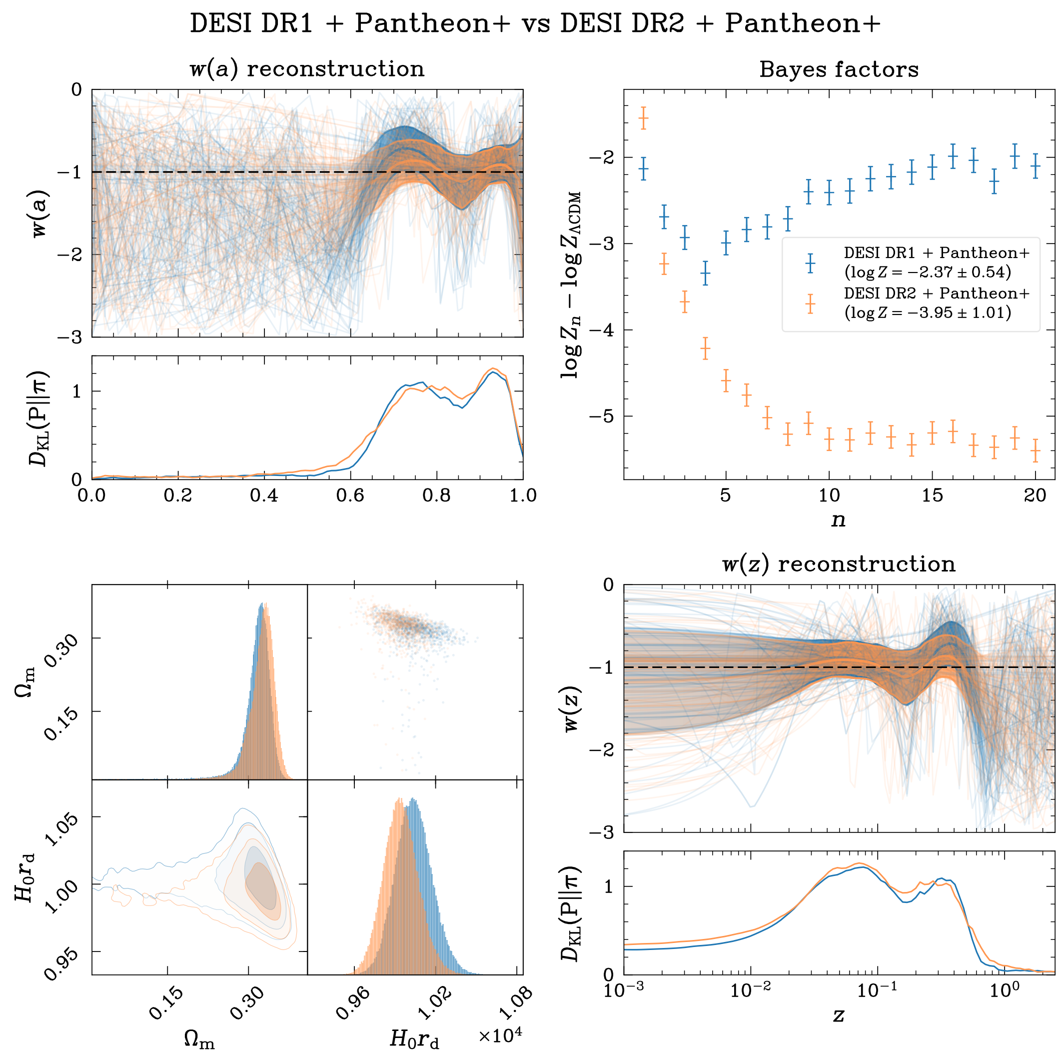
NOW: DESI BAO ALONE

- Start with DESI BAO alone: flexknot reconstruction of $w(a)$ and $w(z)$ (DR2 teal, DR1 black)
- A feature pulls $w(a)$ away from -1 around $z \sim 0.5$ — but only one wiggle
- The evidence still favours ΛCDM at every n (marginalised $\log Z \approx -0.7 \to -1.2$, DR1→DR2)
- DR1→DR2: wCDM ($n{=}1$) and CPL ($n{=}2$) strengthen (CPL preferred), higher-n weaken
- KL-divergence sub-panels show where the BAO constrain w



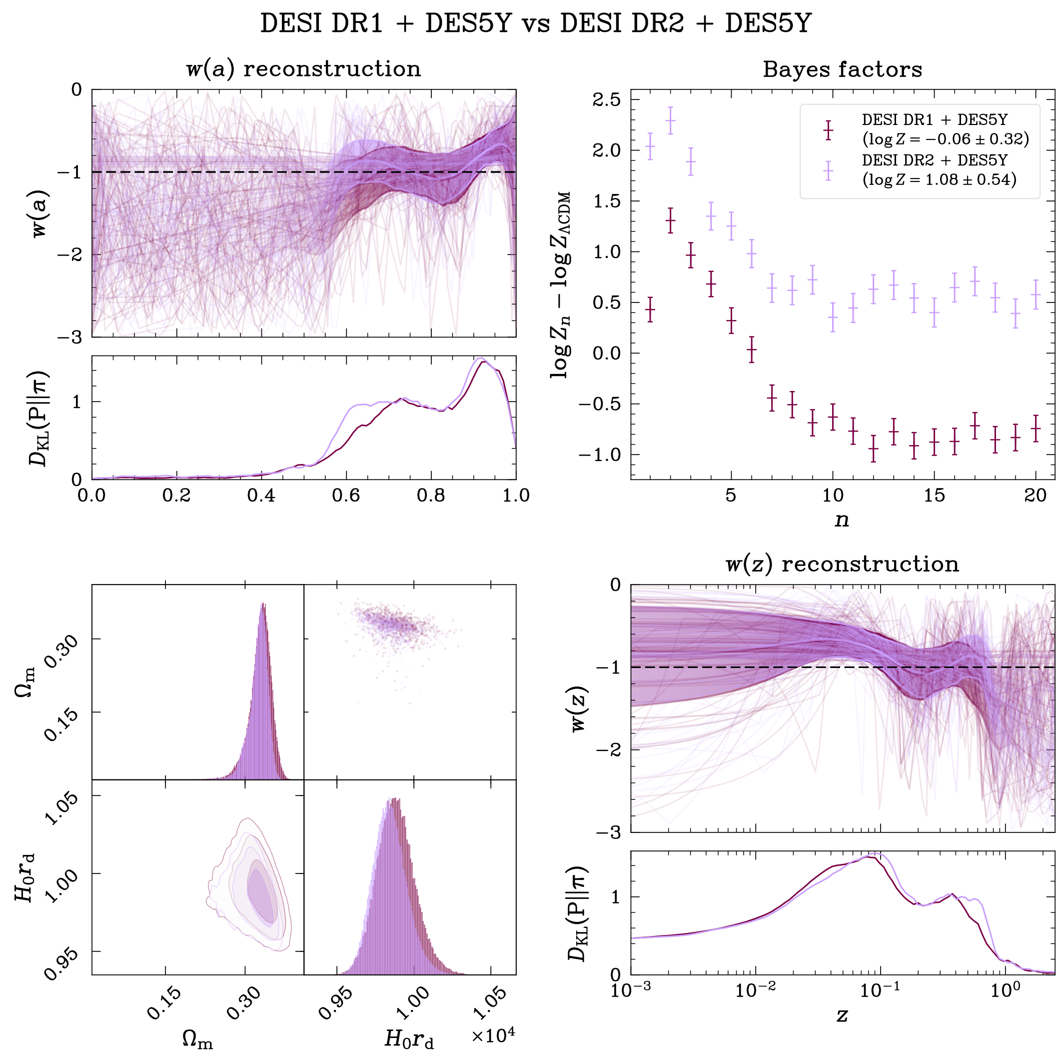
NOW: DESI BAO + PANTHEON+

- Now add Pantheon+ supernovae to DESI BAO (DR2 orange, DR1 blue)
- The free-form reconstruction reveals a "W"-shaped $w(a)$ — too complex for CPL to capture
- Yet the evidence still favours ΛCDM: $\log Z_n - \log Z_{\Lambda\mathrm{CDM}} < 0$ for all n (more so in DR2) — the structure is intriguing, not demanded
- KL-divergence sub-panels show where the data are informative



NOW: DESI BAO + DES-5Y

- Swap in DES-5Y supernovae: now every flexknot model beats ΛCDM
- CPL ($n = 2$) is the most favoured: $\log Z_n - \log Z_{\Lambda\mathrm{CDM}} \approx 2.3$
- Opposite to Pantheon+ (last slide), which favoured ΛCDM — the two SN samples disagree
- So: new physics, or a systematic in the supernovae?



NOW: DYNAMIC, OR SYSTEMATIC?

- Add one nuisance parameter: a magnitude offset Δm_B on the low-z ($z < 0.1$) supernovae external to DES

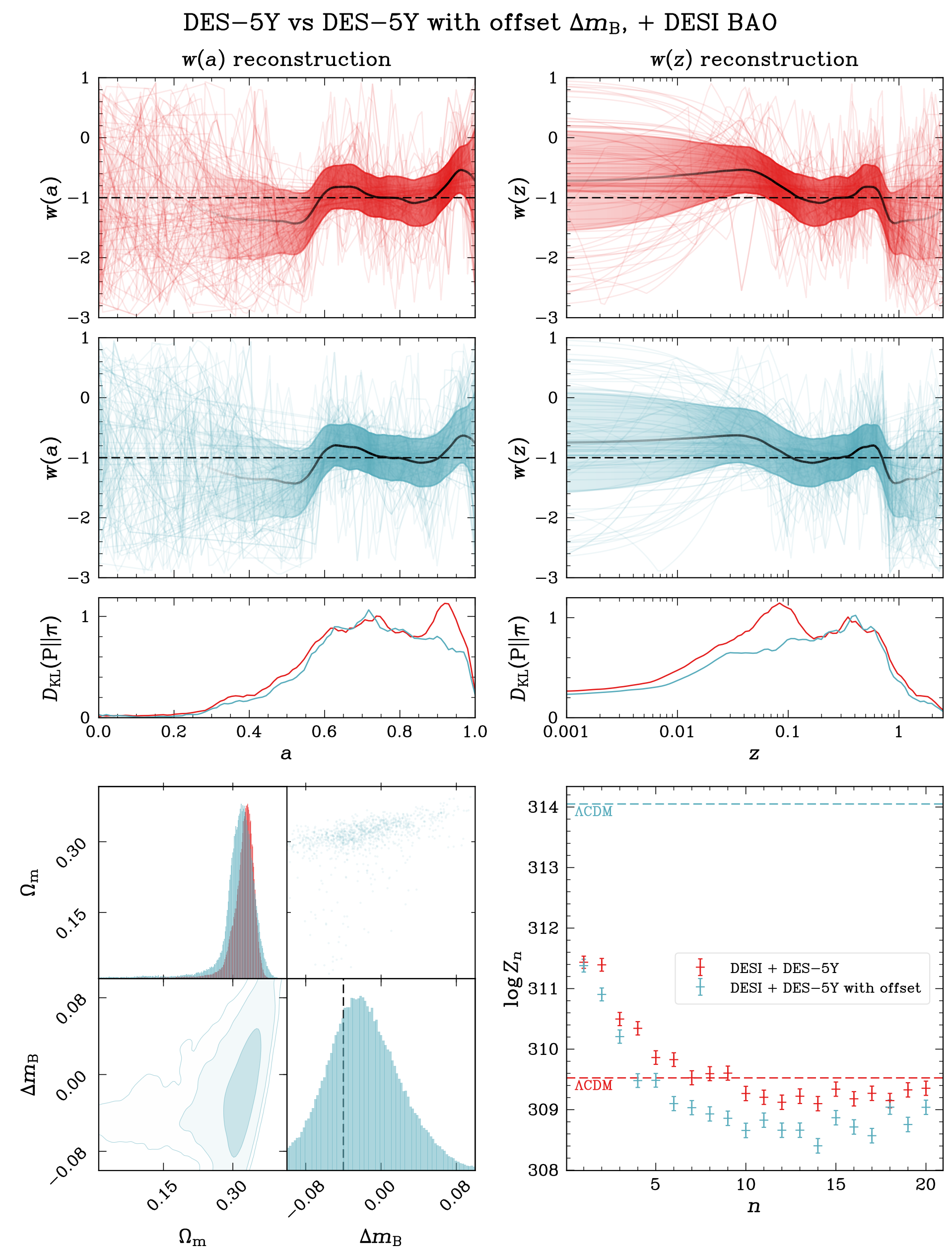
- Let the evidence decide whether the data want it

- Offset ΛCDM beats dynamical dark energy: log-Bayes factor ≈ 4.1 ("decisive") for DESI+DES-5Y

- The offset also significantly reduces the DESI–DES-5Y tension

- Lower-right: the offset ΛCDM evidence (top dashed line) sits far above every flex-knot

- GPU runs (JAX; CPL/ΛCDM): ~ 1 s vs. tens of minutes on CPU



THEN AND NOW: THE SAME IDEA, TRANSFORMED

	THEN (Jaffe 1996)	NOW (Ormondroyd 2025)
Question	open vs. flat universe	cosmological constant vs. dynamical $w(z)$
Model space	4 fixed models	free-form flexknot, averaged over n
Parameters	0–2	up to $\sim$40, variable dimension
Evidence	Gaussian approx. $+$ grid	nested sampling (PolyChord)
Hardware	pen and paper	GPU, $\sim$1 s per run
Systematics	assumed	modelled & selected by evidence

- Thirty years on, the engine is unchanged: the Bayesian evidence* and Occam's razor

Happy 60th birthday, Andrew!

* Terminology first used in astrophysics and cosmology by Hobson, Bridle & Lahav (2002)... 😉